

C-DAC — ASSIGNMENT 03 ANSWERS

QUESTION 01

Reverse a Number. Read integer in main(). Use static method reverseNumber() with a while loop to reverse digits.

LOGIC

Extract digits using `% 10`, build reversed number using `reverse = reverse \* 10 + digit`, reduce number using `/ 10` inside a `while (num > 0)` loop.

CODE

```
// Assignment 03 - Question 01: Reverse a Number
public class Q01 {
    static int reverseNumber(int num) {
        int reverse = 0;
        while (num > 0) {
            int digit = num % 10;
            reverse = reverse * 10 + digit;
            num = num / 10;
        }
        return reverse;
    }

    public static void main(String[] args) {
        int number = 12345;
        int reversed = reverseNumber(number);

        System.out.println("Original Number: " + number);
        System.out.println("Reversed Number: " + reversed);
    }
}
```

EXPECTED OUTPUT

```
Original Number: 12345
Reversed Number: 54321
```

QUESTION 02

Sum of Digits. Read integer in main(). Use static method sumOfDigits() with a while loop to calculate digit sum.

LOGIC

Extract each digit using `num % 10`, add to running sum `sum += digit`, and remove last digit using `num /= 10` inside a `while` loop.

CODE

```
// Assignment 03 - Question 02: Sum of Digits
public class Q02 {
    static int sumOfDigits(int num) {
        int sum = 0;
        while (num > 0) {
            int digit = num % 10;
            sum += digit;
            num /= 10;
        }
        return sum;
    }

    public static void main(String[] args) {
        int number = 9876;
        int sum = sumOfDigits(number);

        System.out.println("Number: " + number);
        System.out.println("Sum of Digits: " + sum);
    }
}
```

EXPECTED OUTPUT

```
Number: 9876
Sum of Digits: 30
```

QUESTION 03

Multiplication Table. Read number in main(). Use static method printTable() with a for loop to display table from 1 to 10.

LOGIC

Use a `for (int i = 1; i <= 10; i++)` loop to print `num \* i` for each step.

CODE

```
// Assignment 03 - Question 03: Multiplication Table
public class Q03 {
    static void printTable(int num) {
        System.out.println("Multiplication Table of " + num + ":");
        for (int i = 1; i <= 10; i++) {
            System.out.println(num + " x " + i + " = " + (num * i));
        }
    }

    public static void main(String[] args) {
        int number = 7;
        printTable(number);
    }
}
```

EXPECTED OUTPUT

```
Multiplication Table of 7:
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

QUESTION 04

Prime Numbers. Read *n* in *main()*. Use static method *findPrimesUpTo()* to display all prime numbers between 1 and *n*.

LOGIC

Iterate numbers from 1 to *n*. For each number, check if it has no divisors other than 1 and itself using `isPrime()` method.

CODE

```
// Assignment 03 - Question 04: Prime Numbers
public class Q04 {
    static boolean isPrime(int n) {
        if (n <= 1) return false;
        for (int i = 2; i <= Math.sqrt(n); i++) {
            if (n % i == 0) return false;
        }
        return true;
    }

    static void findPrimesUpTo(int n) {
        System.out.println("Prime numbers between 1 and " + n + ":");
        for (int i = 1; i <= n; i++) {
            if (isPrime(i)) {
                System.out.print(i + " ");
            }
        }
        System.out.println();
    }

    public static void main(String[] args) {
        int limit = 30;
        findPrimesUpTo(limit);
    }
}
```

EXPECTED OUTPUT

```
Prime numbers between 1 and 30:
2 3 5 7 11 13 17 19 23 29
```

QUESTION 05

Simple Calculator. Read numbers & operator (+, -, *, /, %) in main(). Use static method calculate() with switch statement.

LOGIC

Pass operands and operator char to static method `calculate()`. Evaluate operator using `switch-case` including `%` modulus operator.

CODE

```
// Assignment 03 - Question 05: Simple Calculator
public class Q05 {
    static double calculate(double a, double b, char op) {
        switch (op) {
            case '+': return a + b;
            case '-': return a - b;
            case '*': return a * b;
            case '/':
                if (b != 0) return a / b;
                else {
                    System.out.println("Division by zero error!");
                    return Double.NaN;
                }
            case '%':
                if (b != 0) return a % b;
                else {
                    System.out.println("Modulo by zero error!");
                    return Double.NaN;
                }
            default:
                System.out.println("Invalid operator!");
                return Double.NaN;
        }
    }

    public static void main(String[] args) {
        double num1 = 20;
        double num2 = 6;
        char op = '%';

        double result = calculate(num1, num2, op);
        System.out.println("Calculation: " + num1 + " " + op + " " + num2 + " = " + result);
    }
}
```

EXPECTED OUTPUT

Calculation: 20.0 % 6.0 = 2.0

QUESTION 06

Menu-Driven Area Calculator. Display menu (1. Circle, 2. Rectangle, 3. Triangle, 4. Exit). Read choice & values in main(), compute area in static method.

LOGIC

Display menu options in `main()`. Based on choice, call `calculateArea()` with relevant shape dimensions (radius, length, breadth, height).

CODE

```
// Assignment 03 - Question 06: Menu-Driven Area Calculator
public class Q06 {
    static double calculateArea(int choice, double r, double l, double b, double h) {
        switch (choice) {
            case 1: return Math.PI * r * r;
            case 2: return l * b;
            case 3: return 0.5 * b * h;
            default: return -1;
        }
    }

    public static void main(String[] args) {
        int choice = 2;
        double radius = 5.0;
        double length = 10.0;
        double breadth = 4.0;
        double height = 6.0;

        System.out.println("--- Menu ---");
        System.out.println("1. Area of Circle");
        System.out.println("2. Area of Rectangle");
        System.out.println("3. Area of Triangle");
        System.out.println("4. Exit");
        System.out.println("Selected Choice: " + choice);

        double area = calculateArea(choice, radius, length, breadth, height);
        if (area != -1) {
            System.out.println("Calculated Area: " + area);
        } else {
            System.out.println("Invalid Choice!");
        }
    }
}
```

EXPECTED OUTPUT

```
--- Menu ---
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Exit
Selected Choice: 2
Calculated Area: 40.0
```

QUESTION 07

Find Maximum and Minimum. Read n integers into array in main(). Pass array to static method findMaxMin() to find max and min elements.

LOGIC

Pass array to `findMaxMin()`. Initialize `max` and `min` to `arr[0]`. Iterate through array and update `max` and `min` dynamically.

CODE

```
// Assignment 03 - Question 07: Find Maximum and Minimum
public class Q07 {
    static int[] findMaxMin(int[] arr) {
        if (arr.length == 0) return new int[]{0, 0};
        int max = arr[0];
        int min = arr[0];

        for (int i = 1; i < arr.length; i++) {
            if (arr[i] > max) max = arr[i];
            if (arr[i] < min) min = arr[i];
        }
        return new int[]{max, min};
    }

    public static void main(String[] args) {
        int[] numbers = {34, 12, 89, 5, 67, 23};

        System.out.print("Array elements: ");
        for (int n : numbers) System.out.print(n + " ");
        System.out.println();

        int[] result = findMaxMin(numbers);
        System.out.println("Maximum Element: " + result[0]);
        System.out.println("Minimum Element: " + result[1]);
    }
}
```

EXPECTED OUTPUT

```
Array elements: 34 12 89 5 67 23
Maximum Element: 89
Minimum Element: 5
```

QUESTION 08

Student Marks Analysis. Read marks into array in main(). Pass array to static method analyzeMarks() to calculate total, average, highest, lowest.

LOGIC

Pass marks array to `analyzeMarks()`. Accumulate total, compute average (total / n), and track highest and lowest values.

CODE

```
// Assignment 03 - Question 08: Student Marks Analysis
public class Q08 {
    static void analyzeMarks(int[] marks) {
        int total = 0;
        int max = marks[0];
        int min = marks[0];

        for (int mark : marks) {
            total += mark;
            if (mark > max) max = mark;
            if (mark < min) min = mark;
        }
        double average = (double) total / marks.length;

        System.out.println("Total Student Marks Analysis:");
        System.out.println("Total Marks: " + total);
        System.out.println("Average Marks: " + average);
        System.out.println("Highest Mark: " + max);
        System.out.println("Lowest Mark: " + min);
    }

    public static void main(String[] args) {
        int[] studentMarks = {78, 85, 92, 64, 88};

        System.out.print("Student Marks: ");
        for (int m : studentMarks) System.out.print(m + " ");
        System.out.println();

        analyzeMarks(studentMarks);
    }
}
```

EXPECTED OUTPUT

```
Student Marks: 78 85 92 64 88
Total Student Marks Analysis:
Total Marks: 407
Average Marks: 81.4
Highest Mark: 92
Lowest Mark: 64
```


QUESTION 09

Search an Element. Read n integers into array & target element in main(). Pass to static method searchElement() to perform linear search.

LOGIC

Pass array and search element to `searchElement()`. Loop through array elements sequentially; return index if matched, else -1.

CODE

```
// Assignment 03 - Question 09: Search an Element
public class Q09 {
    static int searchElement(int[] arr, int target) {
        for (int i = 0; i < arr.length; i++) {
            if (arr[i] == target) {
                return i;
            }
        }
        return -1;
    }

    public static void main(String[] args) {
        int[] array = {15, 28, 42, 9, 63};
        int target = 42;

        System.out.print("Array: ");
        for (int el : array) System.out.print(el + " ");
        System.out.println();

        int position = searchElement(array, target);
        if (position != -1) {
            System.out.println("Element " + target + " found at index position " + position + ".");
        } else {
            System.out.println("Element " + target + " not found in the array.");
        }
    }
}
```

EXPECTED OUTPUT

```
Array: 15 28 42 9 63
Element 42 found at index position 2.
```

QUESTION 10

Count Even and Odd Numbers. Read n integers into array in main(). Pass array to static method countEvenOdd() to count evens and odds.

LOGIC

Pass array to `countEvenOdd()`. Iterate through elements: increment `evenCount` if `num % 2 == 0`, else increment `oddCount`.

CODE

```
// Assignment 03 - Question 10: Count Even and Odd Numbers
public class Q10 {
    static int[] countEvenOdd(int[] arr) {
        int evenCount = 0;
        int oddCount = 0;

        for (int num : arr) {
            if (num % 2 == 0) {
                evenCount++;
            } else {
                oddCount++;
            }
        }
        return new int[]{evenCount, oddCount};
    }

    public static void main(String[] args) {
        int[] numbers = {12, 17, 24, 35, 40, 51, 68};

        System.out.print("Array elements: ");
        for (int n : numbers) System.out.print(n + " ");
        System.out.println();

        int[] counts = countEvenOdd(numbers);
        System.out.println("Count of Even Numbers: " + counts[0]);
        System.out.println("Count of Odd Numbers: " + counts[1]);
    }
}
```

EXPECTED OUTPUT

```
Array elements: 12 17 24 35 40 51 68
Count of Even Numbers: 4
Count of Odd Numbers: 3
```